

Muhammad Ahmed Mohsin

Ph.D. Candidate, Stanford Electrical Engineering

muahmed@stanford.edu | [ahmd-mohsin.github.io](https://github.com/ahmd-mohsin) | buildingalexein.ai

RESEARCH INTERESTS

Preference Optimization and Test-Time Scaling. Designing sample-efficient preference-learning and reinforcement-learning methods for aligning language-model reasoning, and adaptive test-time compute methods that dynamically control inference depth, tool invocation, and verification under strict budgets. I am interested in how post-training and inference-time search jointly shape reasoning, and in principled trade-offs among accuracy, latency, and reliability.

Agentic System Architectures and Coding Agents. Building agentic systems that plan, act, and self-verify over long horizons, spanning multi-agent coordination and the software agents I develop at alexein.ai. Through my work at Amazon AGI on coding agents, I study how agents convert test-time compute into solved tasks by exploring diverse solutions and writing trustworthy verifiers, and how these behaviors can be internalized through supervised and reinforcement post-training.

AI Alignment and Benchmarking. Developing persona-based benchmarks and user-simulation frameworks that evaluate how models behave across diverse populations and interactive settings. I am broadly interested in AI alignment, including calibrated uncertainty and reducing sycophancy, and in measurement methodology that makes alignment progress observable and reproducible.

EDUCATION

Stanford University, Bay Area, California 2024–Present

Ph.D. in Electrical Engineering (LLM Post-Training, Reinforcement Learning, Agentic Architectures)

- **CGPA** 4.0/4.0
- **Advisors** [Dr. Emily Fox](#) and [Dr. John Cioffi](#). Research collaboration with Google DeepMind, Meta MSL, Amazon AGI and Microsoft Core AI.
- **Fellowship** Qualcomm Fellowship.

National University of Sciences & Technology (NUST), Islamabad, Pakistan Aug. 2020 – June 2024

Bachelor of Engineering in Electrical Engineering

- **CGPA** 3.87/4.0 | **Specialization GPA** 4.0/4.0
- **Senior Design Thesis** Reinforcement learning for resource allocation optimization in dynamic environments
- **Rector's Gold Medal** Awarded Rector's Gold Medal for best final year project cohort (2024)

RESEARCH EXPERIENCE

Amazon AGI, Applied Research Scientist June 15, 2026 – September 18, 2026

Advisor: Myeongsoo Kim | Project: Teaching Agents to Code Reliably (ICLR 2027)

- Ran a controlled study of six test-time strategies on a fixed Qwen3-Coder-30B agent over SWE-bench Verified, isolating diversity of exploration and diversity of edits as the skills that convert compute into solved bugs and showing that self-authored verification sits near chance.
- Distilled diverse, reactive search into the policy via weighted supervised fine-tuning (pass@1 31.8 → 35.1, pass@8 46.8 → 51.2), then trained a discriminative self-verifier with GRPO using a mutation-gauntlet reward with step-level, threshold-free credit.

Stanford Artificial Intelligence Laboratory (SAIL), Ph.D. Researcher

September 2025 – Present

Advisor: Prof. Emily Fox

- **Continuous-Utility Direct Preference Optimization (CU-DPO)**. Replaced binary preference labels with continuous utility scores to capture fine-grained reasoning quality, aligning models to prompt-based cognitive strategies through a two-stage strategy-selection and execution-refinement pipeline that lifted strategy-selection accuracy from roughly 35 to 46% up to 68 to 78% across seven base models, with gains transferring out of distribution.
- **General Preference Reinforcement Learning (GPRL)**. Preserved multi-dimensional preference structure through the policy update by computing and independently normalizing per-dimension advantages under a General Preference Model, with a drift monitor that detects single-axis exploitation to resist reward hacking, reaching a 56.5% length-controlled win rate on AlpacaEval 2.0.
- **Coverage Preservation in RL as an Off-Policy Constraint**. Showed that on-policy RLVR (e.g. GRPO) narrows reasoning coverage because the gradient routed to a correct mode scales as $O(p_\theta(M))$, then preserved the recoverability of verified reasoning modes with an off-policy, one-sided, base-anchored support floor enforced by teacher forcing, cutting reasoning-mode collapse from 39% to 2.3% and raising the large- k continued-RL ceiling.

Samsung Semiconductors, Ph.D. Researcher

June 2024 – September 2025

Advisor: Dr. John M. Cioffi | Project: Graph Neural Networks for Accelerating Low-Rank SDP Solvers (JMLR 2026)

- Developed a constraint-graph representation of SDPs with a Graph Attention Network encoder to predict rank trajectories, integrating learned rank schedules into low-rank solvers to eliminate manual heuristics and achieve up to $3\times$ speedups on large-scale benchmarks.

SELECTED PUBLICATIONS

* denotes equal contribution. Author name in **bold**.

LLM Alignment, Preference Optimization, and Reasoning

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| [C1] General Preference Reinforcement Learning | NeurIPS 2026 |
| <i>M. Umer*, M. A. Mohsin*, A. Bilal*, E. Fox</i> | |
| [C2] Continuous-Utility Direct Preference Optimization | EMNLP 2026 |
| <i>M. A. Mohsin, M. Umer, A. Bilal, E. Vitercik, J. M. Cioffi</i> | |
| [C3] Active Alignment with Bayesian General Preference Models | CoLM 2026 |
| <i>M. Umer*, M. A. Mohsin*, A. Bilal, E. Vitercik, J. M. Cioffi</i> | |
| [C4] Sycophancy Disentanglement in LLMs via Reward Decomposition | AAAI 2027 |
| <i>M. A. Mohsin*, A. Bilal*, M. Umer, E. Fox</i> | |
| [C5] Coverage Preservation in RL as an Off-Policy Constraint | In Progress |
| <i>M. A. Mohsin, M. Umer, A. Bilal, J. M. Cioffi, E. Fox</i> | |

Test-Time Scaling and Coding Agents

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| [C6] Teaching Agents to Code Reliably | ICLR 2027 |
| <i>M. A. Mohsin*, M. Kim, K. Ruan, S. Garg, V. Kumar, M. K. Ramanathan</i> | |
| [C7] LoRe-KV: Lookahead Output-Perturbation with Response-Entropy Weighting for KV Cache Eviction | ICLR 2027 |
| <i>A. Bilal, M. A. Mohsin, M. Umer, D. F. Hougen</i> | |
| [C8] Epistemic Uncertainty for Test-Time Discovery | NeurIPS 2026 |
| <i>M. A. Mohsin*, K. Riaz*, M. Umer, A. Bilal, J. M. Cioffi, E. Fox</i> | |

- [C9] S^3 : Stratified Scaling Search for Test-Time in Diffusion Language Models CoLM 2026
A. Bilal, M. A. Mohsin*, M. Umer, D. F. Hougen*
- [C10] What If We Allocate Test-Time Compute Adaptively? ICML 2026
A. Bilal, M. A. Mohsin*, M. Umer, D. F. Hougen, J. M. Cioffi*

Multi-Agent Systems, Benchmarking, and Causal Discovery

- [C11] MatrAIX: Simulating the World with 8.3 Billion Persona Agents Invited to Nature
Xiaomin Li, et al., Muhammad Ahmed Mohsin, et al., Dawn Song
- [C12] MicroVerse: An Instrument for Measuring Self-Authored Identity Drift in Long-Horizon Multi-Agent Language-Model Simulations CoLM 2026
Sky Ng, et al., Muhammad Ahmed Mohsin, et al., Yuexing Hao
- [C13] PersonaEval: Persona-Based User Simulation for Evaluating Interactive Applications CoLM 2026
Yifan Simon Liu, et al., Muhammad Ahmed Mohsin, et al., Xiaomin Li
- [C14] When Should Semantic Priors Enter Partially-Identified Causal Discovery? ICLR 2027
M. A. Mohsin, A. Bilal, S. B. Ather, Tomasso, Valentin, L. de Costa
- [C15] Automated Prompt Optimization Improves Robustness of Medical Benchmarks for LLMsML4H 2025
A. Aali, M. A. Mohsin, V. Bikia, A. Singhvi, S. Bedi, H. Cui, M. A. F. Hernandez, A. Unell, Y. Mai, M. A. Pfeffer, S. Koyejo, R. Daneshjou, E. Alsentzer, C. Potts, A. S. Chaudhari
- [C16] On the Fundamental Limits of Large Models at Scale TMLR 2025
M. A. Mohsin, M. Umer, A. Bilal, Z. Memon, M. I. Qadir, S. Bhattacharya, H. Rizwan, A. R. Gorle, M. Z. Kazmi, A. Mohsin, A. Subhan, M. U. Rafique, Z. He, P. Mehta, J. Han, M. A. Jamshed, D. F. Hougen, J. M. Cioffi

Signal Processing, Wireless, and AI

- [C17] Graph Neural Networks for Accelerating Low-Rank SDP Solvers JMLR 2026
M. A. Mohsin, M. Umer, A. Bilal, J. M. Cioffi, E. Vitercik
- [C18] Neural Gaussian Radio Fields for Channel Estimation KDD 2026
M. A. Mohsin, M. Umer*, A. Bilal, J. M. Cioffi*
- [C19] Continual Learning for Wireless Channel Prediction ICML 2025
M. A. Mohsin, M. Umer, A. Bilal, J. M. Cioffi
- [C20] Task-Aware Distributed Source Coding for Correlated Audio Signals Using Perceptual LossAAAI 2024
M. A. Mohsin, A. Bilal, S. Bhattacharya, J. M. Cioffi
- [C21] PyramidTabNet: Transformer-Based Table Recognition in Image-Based Documents ICDAR 2023
M. Umer, M. A. Mohsin, A. Ul-Hasan, F. Shafait

INVITED TALKS AND TUTORIALS

Tutorial IEEE ICMLCN, Abu Dhabi, UAE.
Tutorial IEEE VTC, Boston, USA.
Panel MatrAIX AI Panel, NeurIPS.

REVIEWER AND GRANTS

Conference Reviewer AAAI, ICDAR, NeurIPS, ICML (Silver Reviewer), ICLR, KDD, CoLM
Journal Reviewer IEEE Transactions on AI, Transactions on Machine Learning Research, Applied Artificial

Intelligence.

Travel Grants IEEE GLOBECOM 2024 (Cape Town, South Africa), IEEE ICC 2025 (Canada), Stanford Conference Travel Grant 2025, IEEE ICC 2026 (Glasgow, Scotland).

Best Paper/Poster Awards 6G Summit Abu Dhabi (Best Poster Nomination), IEEE ICC Canada (Best Workshop Paper Award), IEEE Communications Society Competition (honorary mention), IEEE FIT 2026 Best Main Conference Paper Award.

Technical Program Committee NeurIPS 2025, ICASSP 2026.

Workshop Co-Chair IEEE ICC 2026.

Area Chair ICASSP 2026, NeurIPS 2025.

HONORS AND ACHIEVEMENTS

Fellowship Awarded Stanford Graduate Fellowship, Qualcomm Fellowship

Knight-Hennessy Fellowship Finalist.

President's Medal Third position nationwide in pre-engineering group.

Scholarships HSSC Federal Board Scholarship (PKR 200,000). STEP-ECAT Scholarship for top 10 engineering candidates nationwide (PKR 200,000).